Building the Ogun Catchment SWAT+ Model

Step-by-step in QSWAT+ — delineation to calibration, with the traps marked. Attah Gbubemi David.

What you are building, and why the Ogun

SWAT+ models a **drainage basin**, not an administrative area. The floodwater that reaches Lagos through the lagoon falls as rain upstream and arrives down the Ogun, whose catchment extends north into Ogun and Oyo States. Modelling the Ogun basin *is* modelling Lagos's fluvial flood source.

Property	Value
Model domain	Ogun River basin, outlet at Kara/Akute, above the Lagos Lagoon
Delineated area	19,915 km^2
Structure	23 subbasins • 71 channels • 71 LSUs • 243 HRUs
Coordinate system	EPSG:32631 throughout
Simulation period	2010–2023, first year as warm-up
Calibration target	GEOGLOWS v2 reach **140585128**, monthly discharge
Working folder	`C:\SWAT_Ogun\`
QGIS project	`C:\SWAT_Ogun\OgunModel2\OgunModel2.qgs`

Why GEOGLOWS and not NIHSA: gauged Nigerian discharge was not obtainable within the project period. GEOGLOWS ECMWF modelled discharge is the documented data-scarce alternative (Odusanya et al., 2019). It is free and needs no login.

Software, in this order

Component	Version	Note
QGIS	3.44.11 LTR, 64-bit standalone	**install first**
SWAT+ installer	4.0.1	installs QSWAT+ 4.0, SWAT+ Editor 4.0, engine rev 62
GDAL / OGR CLI	bundled with QGIS	`C:\Program Files\QGIS 3.44.11\bin\`

Enable the plugin: **Plugins ■ Manage and Install Plugins ■ Installed**, tick **QSWATPlus**. An **S+** icon appears in the toolbar.

> **TRAP — Order matters**
> SWAT+ installs a plugin *into* QGIS. Installing SWAT+ first leaves you with an editor and an engine but no toolbar icon.

Step 1 — The DEM

Source: Copernicus GLO-30, from the AWS open bucket, no login. Chosen over SRTM because it needs no Earthdata account and covers the whole basin.

1.1 Download the six tiles covering N06–N08 x E002–E003.

1.2 Mosaic and reproject to EPSG:32631 at 30 m:

```
gdalwarp -t_srs EPSG:32631 -tr 30 30 -r bilinear -of GTiff -co COMPRESS=LZW \
  cop_*.tif  ogun_dem_utm31.tif
```

1.3 Resample to 90 m for delineation:

```
gdalwarp -tr 90 90 -r bilinear ogun_dem_utm31.tif ogun_dem_90m_utm31.tif
```

1.4 Clip to the basin window — this is the DEM you delineate with:

```
gdalwarp -t_srs EPSG:32631 -te 2.6 6.55 3.9 9.1 -te_srs EPSG:4326 \
  -tr 90 90 -r bilinear ogun_dem_utm31.tif ogun_dem_90m_clip.tif
```

Expect: `ogun_dem_90m_clip.tif`*, 1597 x 3134 px, about 19 MB, 5 million cells. Elevation 0–585 m, 100% valid.*

TRAP — 30 m is too fine for a 23,000 km2 basin
The first attempt used the 30 m DEM — 81 million cells. TauDEM's `D8FlowDir.exe` ran for **3.4 hours** with no progress and QGIS showed "Not Responding". Diagnosed by watching `Watershed\Rasters\DEM\` for new files and checking CPU time with `Get-Process`. Match DEM resolution to basin size: 90 m is standard at this scale and runs in minutes.

TRAP — Clip before delineating
The full 90 m rectangle still included a large block of flat ocean, lagoon and coastal plain *south* of the outlet. Flat ground with no drainage direction stalled D8FlowDir and crashed the machine. The basin drains entirely from north of the outlet, so clipping loses nothing.

Step 2 — Burn-in rivers and the outlet point

Why: on terrain this flat, D8 routing disperses flow instead of concentrating it, and no integrated main channel forms. Burning in a known river network forces the water to follow real channels.

2.1 Extract HydroRIVERS, clip to the basin window and reproject:

```
ogr2ogr -f "ESRI Shapefile" -spat 2.6 6.55 3.9 9.1 -t_srs EPSG:32631 \
  ogun_rivers_burn.shp  HydroRIVERS_v10_af.shp
```

2,701 reaches. HydroRIVERS independently confirms a maximum upland area of 24,816 km^2 in the basin, which is the cross-check that the extent is right.

2.2 Create the outlet shapefile at the true main stem near Kara/Akute. Required attributes: `ID=1, INLET=0, RES=0, PTSOURCE=0`. Save as `ogun_outlet2.shp`.

The corrected outlet was snapped to the nearest HydroRIVERS main-stem vertex with upland area above 3,000 km^2. It drains 22,007 km^2.

TRAP — The first outlet was 2.3 km off the main stem
With the original outlet on the raw clipped DEM, maximum flow accumulation was only **462 km^2** against the Ogun's ~23,000, and QSWAT+ warned it could find only one stream and one subbasin. The DEM was verified clean — 100% valid, 0–585 m — so this was a routing problem, not a data gap. **Burn-in plus the corrected outlet took flow accumulation to 20,592 km^2.** This is the single most important fix in the whole build.

Step 3 — Delineate the watershed

QGIS → the **S+** icon → **New Project** → parent `C:\SWAT_Ogun`, name `OgunModel2`, accept the default SWAT+ database. Then **Delineate Watershed**.

Setting	Value	Why
DEM	`ogun_dem_90m_clip.tif`	30 m hung TauDEM for 3.4 h
Burn in existing stream network	**ON** → `ogun_rivers_burn.shp`	flat terrain will not route without it
Use an inlets/outlets shapefile	**ON** → `ogun_outlet2.shp`	the corrected main-stem outlet
Snap threshold	500 m	lets QSWAT+ find the stream cell
Channel threshold	**150 km^2**	the default 40 produced degenerate micro-channels — see the trap below

Setting	Value	Why
Number of processes	**0**	single-process; avoids MPI/TauDEM errors

Then, in order: **Create streams** → **Review snapped** (confirm one outlet, on a stream) → **Create watershed** → **OK**.

> ***Expect: 23 subbasins, 19,915 km^2, 71 channels.*** *Outputs in* `OgunModel2\Watershed\Shapes\`. *Extent reaches north to about 8.9°N.*

> **TRAP — Channel threshold 40 breaks the HRU step later**
> At the default threshold, burn-in leaves degenerate micro-channels: 277 channels against 276 LSUs. The mismatch only surfaces two steps later, as `Internal error: Id 490 has not been added to table gis_lsus`. Raising the threshold to 150 km^2 gives 71 well-formed channels and the HRUs then build cleanly. Set it correctly now rather than debugging it later.

Step 4 — Land use, soil and HRUs

Land use

ESA WorldCover 10 m, mosaicked and resampled to 90 m with `-r mode` (majority) at the basin extent → `landuse\ogun_landuse_90m.tif`. Basin cover: 43% tree, 23% shrub, 16% crop, 12% grass, 4.5% built.

WorldCover	10	20	30	40	50	60	80	90	95
SWAT+ code	frst	rngb	rnge	agrl	urml	bsvg	watr	wetl	wetf

Soil

SoilGrids 250 m sand and clay, 0–5 cm, pulled over the basin window via `/vsicurl/` with no login, converted to percentages and classified into the 12 USDA texture classes → `soil\ogun_soil_90m.tif`. Basin: **88% sandy loam, 11% sandy clay loam**.

SWAT+ ships FAO soils that do not usefully cover this basin, so a custom database was built by cloning the FAO schema into `OgunModel2.sqlite`: `texture_usersoil` (12 rows, representative HYDGRP / BD / AWC / K / CBN / clay-silt-sand per texture) and `texture_soils` (SOIL_ID to SNAM).

Single-layer representative soils, using topsoil texture, is a documented data-scarce simplification — state it rather than let it be found.

Create HRUs

Field	Value
Landuse map	`ogun_landuse_90m.tif`
Landuse lookup	**table** `worldcover_landuses`
Soil map	`ogun_soil_90m.tif`
Soil data	**usersoil**
Soil lookup / usersoil	`texture_soils` / `texture_usersoil`
Slope bands	single band `[0, 9999]`
Method	Read → **Filter by landuse, soil, slope** → Percent
Thresholds	Landuse 10% / Soil 10% (click **Go** after each)

> ***Expect: 23 subbasins, 71 channels, 71 LSUs, 243 HRUs****, filtered down from 1,971.*

TRAP — The landuse lookup must be a database table, not a CSV
QSWAT+ will accept a CSV path and then fail. Clone the `Example1_landuses` schema (columns LANDUSE_ID, SWAT_CODE) into the project database as `worldcover_landuses` and point the dialog at the table.

Step 5 — Weather inputs

Google Earth Engine (free, noncommercial signup). Run `swat\gee_weather.js` in the Code Editor: daily **CHIRPS** rainfall and **ERA5-Land** tmax/tmin, 2010–2023, sampled at the 23 subbasin centroids, exported to Drive.

Then `build_weather.py` converts them to SWAT+ format in `weather\climate\`: 23 `.pcp` + 23 `.tmp` files plus `pcp.cli` and `tmp.cli`. Station elevations are sampled from the DEM; `nbyr=14, tstep=0`.

Expect: *the bimodal south-west Nigeria signal with the August break. Station 23 in 2020: June 371 mm, August 16 mm, September 243 mm. Daily range 0–92 mm.*

TRAP — Single-band CHIRPS returns a null column in Earth Engine
`reduceRegions` on a one-band image produces an unnamed null column. Fix with `ee.Reducer.mean().setOutputs(['pcp'])` and `unmask(0)`. Two-band temperature works first time, which is what makes the rainfall failure confusing.

Step 6 — First run and baseline

In SWAT+ Editor: load the weather generator from the global `swatplus_wgn.sqlite` table `wgn_cfsr_world`, tick **using observed weather data**, and point at the 23 observed stations. Solar, wind and humidity stay as `sim`. Simulation 2010–2023 with `nyskip=1`.

TRAP — Monthly output needs a manual edit that is easily lost
The default print setting is yearly. Edit `print.prt` to set `channel` and `channel_sd` to monthly, then re-run with **Write input files UNCHECKED** — otherwise the editor regenerates print.prt and silently discards the edit.

Baseline, uncalibrated, 2011–2023	Value
Simulated mean	337 m^3/s
Observed mean (GEOGLOWS)	226 m^3/s
NSE	–0.10
R^2	0.51
PBIAS	–48.9%

Read that diagnosis properly. R^2 around 0.5 means the seasonal *shape* is already right; the *partitioning* is wrong. Wet-season peaks are too high (excess surface runoff) and dry-season baseflow is too weak (January–March simulated ~4 against observed ~50). That tells you exactly which parameters to reach for.

Step 7 — Calibration

Driven by `C:\SWAT_Ogun\calib.py`, which writes `calibration.cal`, registers it in the `file.cio` “chg” slot, runs the engine directly, and scores NSE / R^2 / PBIAS against GEOGLOWS monthly at outlet `cha01`. No GUI.

Parameter	Value	What it fixed
perco	0.30	flow is baseflow-dominated; the default 0.90 drained soil water before it could evaporate. First big lever: NSE –0.10 → 0.18
petco	1.20	the real ET lever. Fixed the volume: PBIAS –30% → +3%

Parameter	Value	What it fixed
canmx	12	canopy interception; trimmed the over-high wet-season peaks
alpha	0.022	slows groundwater release so dry-season flow does not vanish
flo_min	10	aquifer holds a reserve before releasing
bf_max	0.55	caps the baseflow rate

esco, epco, awc and revap were inert or backwards in rev 62 — they raised water rather than lowering it. Abandoned rather than forced.

Period	NSE	R^2	PBIAS
Calibration 2011–17	0.41	0.56	+6.1%
Validation 2018–23	**0.57**	**0.64**	–3.2%
Full 2011–23	**0.507**	**0.606**	**+1.6%**

Expect: *all thresholds met (NSE ≥ 0.50, R^2 ≥ 0.60, \|PBIAS\| ≤ 25%), and* ***validation scores higher than calibration*** *— an overfitted model does the opposite. Mean simulated 222 against observed 226 m^3/s.*

TRAP — The margins are thin — re-check after any change
NSE 0.507 against a 0.50 threshold, R^2 0.606 against 0.60. Any change to forcing or parameters can drop the model below the thesis criteria. Re-run `4_recheck_calibration.bat` rather than trusting a recorded number.

Step 8 — Rainfall scenarios

`scenario.py` scales all 23 `.pcp` files, runs the calibrated model, then restores the originals.

Rainfall	Mean m^3/s	Change in mean	Peak month	Change in peak
Baseline	222	—	1420	—
+10%	257	**+15.5%**	1558	+9.7%
+20%	292	+31.3%	1686	+18.7%
+30%	328	+47.6%	1821	+28.2%

Point at the +10% row. A 10% rainfall increase produces a 15.5% flow increase. That non-linearity is the entire justification for building a process-based model alongside the AHP index — a weighted linear index can only ever respond linearly. Once the soil saturates, extra rain has nowhere to go but the river.

The wettest simulated year is 2019 (mean 308, peak 1053). **2022 shows near-baseline** in a rainfall-only simulation, because the 2022 flood was driven by Oyan Dam releases. Record that as a limitation before anyone raises it.

Step 9 — Independent validation

Calibration and validation both used GEOGLOWS, split only by period, so nothing outside GEOGLOWS had ever tested the model. Two independent checks close that gap.

Reference	Monthly NSE	R^2	PBIAS	Annual mm/yr
MODIS MOD16A2GF (satellite)	0.14	0.36	+14.4%	920
TerraClimate aet (corroborating)	0.18	0.39	+12.5%	900
SWAT+ simulated	—	—	—	**787**

The two references agree with each other (920 against 900 mm/yr), which is what makes this robust rather than an artefact of one product.

Verdict: partial pass. Volume sits inside the ±25% Moriasi band. Seasonal dynamics clearly fail — SWAT+ swings only about 26 mm across the year against MODIS's 86 mm, so the amplitude is roughly **three times too flat**. State this before anyone else finds it.

> **TRAP — Sample the basin outline, not a bounding box**
> The first MOD16 export used a rectangle around the catchment: 40,397 km^2 against the basin's 19,915, so roughly half of every sampled pixel lay outside the watershed. Fixed by inlining the dissolved subbasin outline into the GEE script. **The correction changed no conclusion** (920 against 935 mm/yr, NSE 0.14 against 0.09) — which is worth saying: the finding survives a two-fold change in sampling area, so it is a property of the model, not of how the region was drawn.

Why the ET was not improved — have this ready

petco	Discharge NSE / R^2 / PBIAS	ET NSE / PBIAS	Meets thresholds
1.20 (retained)	0.507 / 0.606 / +1.6%	0.180 / +12.5%	**yes**
1.35	0.530 / 0.596 / +15.7%	0.303 / +6.6%	no — R^2 0.596
1.50	0.507 / 0.583 / +28.7%	0.367 / +1.1%	no — R^2 and PBIAS

ET improves monotonically as discharge degrades — the objectives genuinely conflict. Only the incumbent meets all three criteria, so the incumbent is retained. `petco` is already pinned at its permitted maximum in `cal_parms.cal`; reaching 1.35 required temporarily raising that ceiling, which was **restored afterwards**. Two unrelated satellite products agree the model wants more ET than the parameter range allows.

Every trap in one list

Step	What goes wrong	The fix
1	30 m DEM hangs TauDEM for hours	resample to 90 m
1	flat coastal block crashes D8FlowDir	clip to the basin window
2	flow accumulation caps at 462 km^2	burn in HydroRIVERS and correct the outlet
3	channel threshold 40 makes micro-channels	raise to 150 km^2
3	MPI/TauDEM errors	set processes to 0
4	“Id 490 not in gis_lsus”	caused by step 3; re-delineate
4	landuse lookup rejected	it must be a DB table, not a CSV
5	CHIRPS returns a null column	setOutputs(['pcp']) and unmask(0)
6	monthly output reverts to yearly	re-run with Write input files unchecked
7	esco / epco / awc behave backwards	rev-62 quirk; use perco and petco instead
9	ET sampled over a rectangle	inline the dissolved basin outline